

Model-Based Reinforcement Learning for Quadruped Locomotion: A Synthetic Study

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Abstract

Quadruped robots must adapt learned locomotion policies to hardware that differs from simulation. This synthetic study presents a model-based reinforcement learning pipeline for the TerraHound-4 platform, centred on the Zephyr-7 adaptation scheduler. Zephyr-7 reduces sim-to-real adaptation time by 38% relative to a fixed-schedule baseline while retaining 96% of asymptotic reward.

1 Introduction

Legged locomotion is a convenient testbed for model-based reinforcement learning because dynamics are continuous, contact-rich, and partially observable. Simulation provides cheap experience, but the sim-to-real gap means that adaptation after deployment dominates total training cost. This study is organized as follows. Section 2 describes the TerraHound-4 platform and policy architecture. Section 3 covers world-model training. Section 4 introduces the Zephyr-7 scheduler, Section 5 reports locomotion results, and Sections 6 and 7 discuss failure modes and conclusions.

2 The TerraHound-4 Platform and Policy Architecture

TerraHound-4 is a synthetic twelve-actuator quadruped with a nominal mass of 18 kg and a top speed of 3.2 m/s. Its simulator exposes randomized friction, actuator lag, and payload parameters; the physical platform is assumed to differ from the simulator along exactly these axes, which makes the sim-to-real gap controllable in our protocol.

The policy is a six-layer Transformer encoder with learned absolute positional embeddings over 256-step proprioceptive windows, followed by a linear action head. We deliberately use learned positional embeddings rather than rotary variants because the window length is fixed at deployment, so long-context extrapolation is unnecessary and the learned table is the simpler option.

3 World-Model Training

The world model predicts next-window proprioception and foot contacts from action-conditioned history. It is trained on 40 million simulated steps with randomized dynamics, then frozen. Planning uses short-horizon shooting with the learned model rather than learned value expansion, which we found more stable under contact discontinuities.

4 The Zephyr-7 Adaptation Scheduler

Adaptation after deployment alternates between collecting real rollouts and fine-tuning the world model and policy. The Zephyr-7 scheduler allocates these phases adaptively: it triggers a new collection phase only when model disagreement on the current replay buffer exceeds a calibrated threshold, and it lengthens fine-tuning phases geometrically as disagreement decays. Compared with a fixed alternating schedule, Zephyr-7 reduces sim-to-real adaptation time by 38% on TerraHound-4, from a median of 14.2 hours to 8.8 hours of onboard compute.

The scheduler adds two hyperparameters, the disagreement threshold and the growth factor, and no additional models. Ablation with a disagreement-triggered but constant fine-tuning length recovers only half of the saving, indicating that phase lengthening contributes as much as triggering.

5 Locomotion Results

After adaptation, policies retain 96% of their asymptotic simulated reward on flat terrain, 91% on randomized gravel, and 84% on wet low-friction surfaces. Emergency-stop latency after payload shifts of 3 kg remains within 120 ms for all Zephyr-7 runs, matching the fixed-schedule baseline.

6 Failure Modes

Three failure modes dominate. First, disagreement calibration drifts when actuator lag exceeds the randomized range, causing under-collection. Second, the geometric phase lengthening occasionally locks in a suboptimal schedule after early lucky rollouts. Third, the learned positional table prevents window-length changes after deployment, so any control-frequency change requires retraining the policy head.

7 Conclusions

Adaptive scheduling of real-world adaptation is the largest single lever in our sim-to-real budget. The Zephyr-7 scheduler cuts adaptation time by 38% on TerraHound-4 without extra models, while a fixed-window Transformer policy with learned positional embeddings suffices when deployment context length is constant. Future work targets online recalibration of disagreement thresholds.

References

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